Gradient Boosting and Next-Day Equity Direction: A Walk-Forward Test with Cross-Asset Features

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the out-of-sample ROC AUC of the gradient-boosting classifier for the next-day S&P 500 direction equals 0.5; H_1: it is larger
Data: 1,963 out-of-sample sessions, 2019-01-02 to 2026-10-22, 12 cross-asset features published by the close
Method: Yearly walk-forward folds with a 5-day embargo, fixed hyper-parameters, logistic-regression baseline, 5,000-draw label-permutation test
Results: GBM AUC 0.5068 (permutation $p = 0.3081$), logistic AUC 0.4894; accuracy 53.1% versus 54.7% for always predicting the majority class.
Research decision: REJECTED

1 Design

The target is the sign of the next-day S&P 500 return, a task for which return predictability is known to be weak [Gu et al., 2020]. Features use only information published by the close of day t; each yearly fold trains on earlier years with an embargo of five sessions before the test year. Hyper-parameters are fixed before the evaluation, and nothing is tuned on out-of-sample data.

2 Results

A long-short rule that follows the predicted sign has an annualised Sharpe ratio of 0.47 before costs (descriptive only).

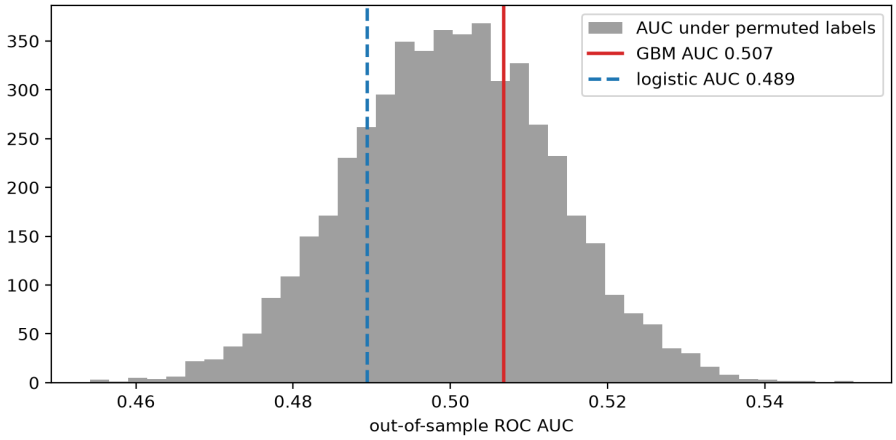


Figure 1: Out-of-sample AUC against the permutation distribution.

3 Limitations

Permuting labels treats daily directions as exchangeable, which is a good approximation for daily index returns but ignores volatility clustering. An AUC above 0.5 does not imply a tradable edge after costs. All features come from Yahoo Finance and FRED closes that are not perfectly synchronous.

References

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.